

Multi-Source Candidate Data Transformer

Technical Design – Eightfold Engineering Intern Assignment

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Pipeline Overview

The pipeline follows a modular, seven-stage sequential architecture. Each stage is an isolated function, keeping the system deterministic and explainable end-to-end:

Detect → **Extract** → **Normalize** → **Merge** → **Confidence** → **Project** → **Validate**

- **Detect** – Identifies source type by file extension or structure (CSV, ATS JSON, LinkedIn JSON, GitHub JSON, free-text notes).
- **Extract** – Parses raw fields into an intermediate dict per source – direct column mapping for CSV, schema mapping for each JSON API (ATS / LinkedIn / GitHub), and regex extraction (look-aheads and exclusion patterns) for free-text notes.
- **Normalize** – Standardizes formats independent of source: phones to E.164, dates to YYYY-MM, skill names into a canonical vocabulary via a lookup dict.
- **Merge** – Matches records across sources by normalized full_name (case-insensitive, whitespace-trimmed) – the only reliable join key, since structured and unstructured sources share no common ID.
- **Confidence** – Scores each field via a fixed, deterministic rule engine, then averages populated field scores into overall_confidence.
- **Project** – Reshapes the internal canonical record into the requested output via a generic, configuration-driven function – never special-cased per request.
- **Validate** – Strictly type-checks and regex-validates the projected output against the active schema before it is returned.

Canonical Schema & Normalization

The internal record uses the canonical schema as given – (*candidate_id*, *full_name*, *emails*[], *phones*[], *location*, *links*, *headline*, *years_experience*, *skills*[], *experience*[], *education*[], *provenance*[], *overall_confidence*) – without modification, since it already separates identity, contact, and professional-history concerns cleanly. **Phones** normalize strictly to E.164; as this pipeline targets Indian candidate datasets, ambiguous 10-digit numbers default to the **+91** country code. **Skills** normalize via a canonical alias dictionary (e.g. “ML” → “Machine Learning”) to collapse variants across resumes and ATS systems; each skill carries its own confidence and a sources[] list, scored by the same agreement rule as other fields, rather than a single flat list. **Location** normalizes to {city, region, country}, with country mapped to its ISO-3166 alpha-2 code (e.g. “Bengaluru” → country: “IN”) instead of a free-text string. **Links** are extracted rigorously even from unstructured text – portfolio URLs lacking an https:// prefix are safely caught and rebuilt rather than dropped.

Merge & Confidence Policy

Match key: normalized full_name. **Conflict rule:** when sources disagree, the structured source (ATS JSON or CSV) wins over a value inferred from prose, since it is a labeled field rather than a regex guess – but both values are retained in provenance with the losing value marked, so a human reviewer can audit the decision rather than see only the winner.

Single structured source	0.70
Single unstructured source (regex)	0.50
Two sources, values agree	0.95
Two sources, values disagree	0.40 (provenance records the trail)
Missing from all sources	null – excluded from average

overall_confidence is the mean of all non-null field confidence scores for a candidate. This formula intentionally rewards cross-source agreement and penalizes both extraction uncertainty and conflict – directly reflecting the principle that wrong-but-confident is worse than honestly-empty.

Runtime Config / Projection Layer

The canonical record and the output schema are kept strictly separate: *project(record, config) -> output*. The config controls field subset selection and renaming via a from path (e.g. “primary_email” from emails[0]), per-field normalization overrides (e.g. normalize: E164), toggling of confidence/provenance inclusion, and missing-value behavior (null / omit / error). Because projection is a single generic function driven entirely by config data, new output shapes require zero code changes – only a new config file.

Edge Cases Handled (and Deliberate Scope Cuts)

- Missing source file: pipeline logs a warning and continues with whatever sources are present; never crashes on an absent file.
- Malformed/empty CSV row or JSON object: skipped individually with a logged warning; does not abort the batch.
- Cross-source value conflict: resolved via the merge policy above; both values preserved in provenance.
- Un-prefixed URLs in free text: portfolio links lacking http:// or www. are extracted via negative-lookbehind regex to avoid confusion with email domains.
- Unstructured note with no extractable phone/email/skills: fields remain null – never fabricated.

Out of scope (time-boxed): fuzzy name matching (exact match only); multi-language extraction.